

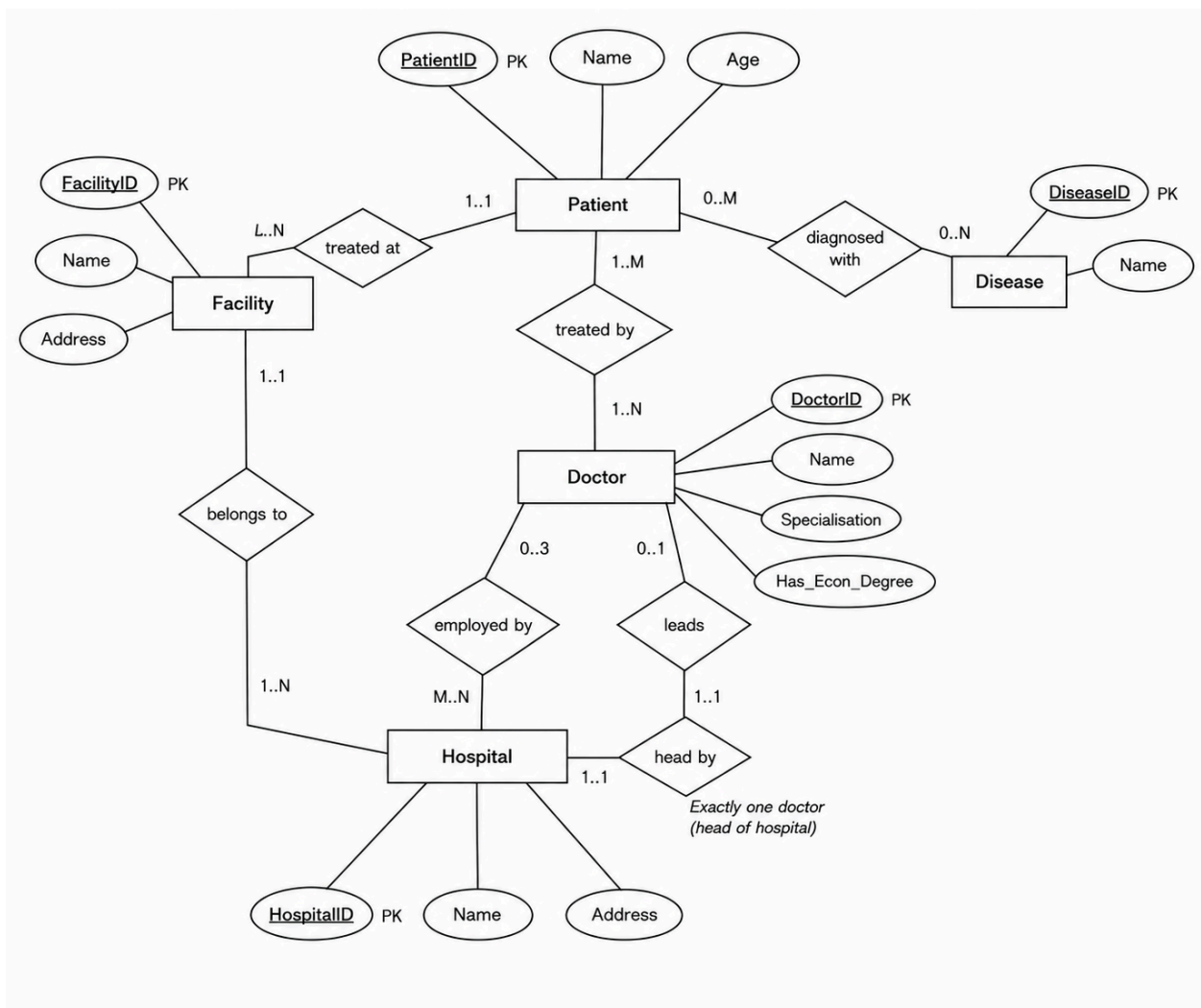
DBMS ASSIGNMENT 2:

ER Model and ER-to-Relational Mapping

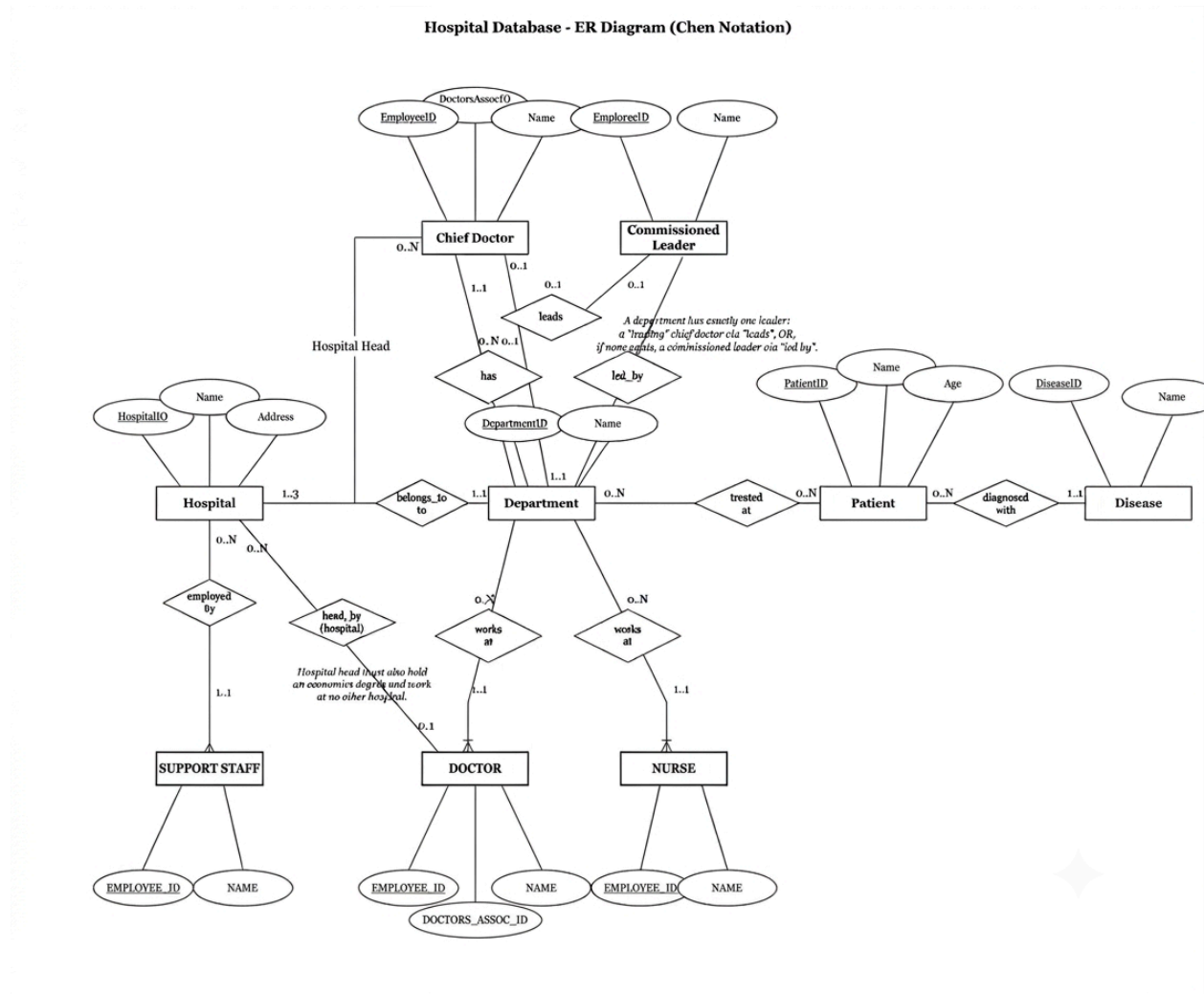
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Question 1



Question 2



Question 3

An arrow from a relationship diamond into an entity means that entity is on the "one" side of that relationship (functional: each instance of the *other* entity participates in at most one instance of this one). Since every relationship here (P, Q, R, T, S, U) is therefore 1:N (no M:N relationships), no separate relation is needed for any relationship — each merges as a foreign key into the entity on the "many" side. This gives the minimum number of relations: one relation per entity, four relations total.

Cardinality reading:

- P: arrow → B, so B is "1", A is "N" → FK goes in A
- Q: arrow → C, so C is "1", B is "N" → FK (+ attribute B2) goes in B
- R: arrow → D, so D is "1", C is "N" → FK goes in C
- T, S, U: arrow → A (each), so A is "1", D is "N" in each → three separate FKs go in D

Relational Schema:

A(**A1**, A2, B_fk)
B(**B1**, B2, C_fk)
C(**C1**, C2, D_fk)
D(**D1**, D2, D3, A_fk_T, A_fk_S, A_fk_U)

Primary Keys:

- A: A1
- B: B1
- C: C1
- D: D1

Foreign Key References:

- A(B_fk) references B(B1) : from relationship P
- B(C_fk) references C(C1) : from relationship Q
- C(D_fk) references D(D1) : from relationship R
- D(A_fk_T) references A(A1) : from relationship T
- D(A_fk_S) references A(A1) : from relationship S
- D(A_fk_U) references A(A1) : from relationship U

T, S, U are three distinct 1:N relationships between the same pair of entities (A and D), so D needs three separate foreign key columns, one per relationship — they can't be combined into a single FK.

Question 4

For Question 1

Part (a): Straightforward Transformation (one relation per relationship type)

Entity relations:

Patient(**PatientID**, Name, Age)
Facility(**FacilityID**, Name, Address)
Disease(**DiseaseID**, Name)
Doctor(**DoctorID**, Name, Specialisation, Has_Econ_Degree)
Hospital(**HospitalID**, Name, Address)

Relationship relations:

TreatedAt(**PatientID**, FacilityID)
TreatedBy(**PatientID**, **DoctorID**)
DiagnosedWith(**PatientID**, **DiseaseID**)

BelongsTo(**FacilityID**, HospitalID)
EmployedBy(**DoctorID**, **HospitalID**)
Leads(**HospitalID**, DoctorID)

Foreign keys:

- TreatedAt.PatientID → Patient(PatientID); TreatedAt.FacilityID → Facility(FacilityID)
- TreatedBy.PatientID → Patient(PatientID); TreatedBy.DoctorID → Doctor(DoctorID)
- DiagnosedWith.PatientID → Patient(PatientID); DiagnosedWith.DiseaseID → Disease(DiseaseID)
- BelongsTo.FacilityID → Facility(FacilityID); BelongsTo.HospitalID → Hospital(HospitalID)
- EmployedBy.DoctorID → Doctor(DoctorID); EmployedBy.HospitalID → Hospital(HospitalID)
- Leads.HospitalID → Hospital(HospitalID); Leads.DoctorID → Doctor(DoctorID)

Total: 11 relations (5 entity relations + 6 relationship relations)

Part (b): Minimized Transformation (fewest relations)

Only genuinely many-to-many relationships (**TreatedBy**, **DiagnosedWith**, **EmployedBy**) require their own relation. All 1:N and 1:1 relationships are merged as foreign keys into the entity on the "one" side.

Resulting Schema:

Patient(**PatientID**, Name, Age, FacilityID)
Facility(**FacilityID**, Name, Address, HospitalID)
Disease(**DiseaseID**, Name)
Doctor(**DoctorID**, Name, Specialisation, Has_Econ_Degree)
Hospital(**HospitalID**, Name, Address, HeadDoctorID)
TreatedBy(**PatientID**, **DoctorID**)
DiagnosedWith(**PatientID**, **DiseaseID**)
EmployedBy(**DoctorID**, **HospitalID**)

Primary Keys:

- Patient: PatientID
- Facility: FacilityID
- Disease: DiseaseID
- Doctor: DoctorID
- Hospital: HospitalID
- TreatedBy: (PatientID, DoctorID)
- DiagnosedWith: (PatientID, DiseaseID)
- EmployedBy: (DoctorID, HospitalID)

Foreign Keys:

- Patient.FacilityID $\rightarrow$ Facility(FacilityID) : from "treated at"; NOT NULL since every patient treated at exactly one facility
- Facility.HospitalID $\rightarrow$ Hospital(HospitalID) : from "belongs to"; NOT NULL since every facility belongs to exactly one hospital
- Hospital.HeadDoctorID $\rightarrow$ Doctor(DoctorID) : from "leads"; NOT NULL since every hospital has exactly one director
- TreatedBy.PatientID $\rightarrow$ Patient(PatientID); TreatedBy.DoctorID $\rightarrow$ Doctor(DoctorID)
- DiagnosedWith.PatientID $\rightarrow$ Patient(PatientID); DiagnosedWith.DiseaseID $\rightarrow$ Disease(DiseaseID)
- EmployedBy.DoctorID $\rightarrow$ Doctor(DoctorID); EmployedBy.HospitalID $\rightarrow$ Hospital(HospitalID)

Total: 8 relations (down from 11 in part (a)) — the three M:N relationships (TreatedBy, DiagnosedWith, EmployedBy) cannot be avoided since neither participating entity has a maximum cardinality of 1, so no foreign-key merge is possible for these three.

For Question 2

Part (a): Naive transformation (every relationship type gets its own relation)

Entity relations (9):

1. **Hospital**(HospitalID, Name, Address)
Primary key: HospitalID
2. **Department**(DepartmentID, Name)
Primary key: DepartmentID
3. **ChiefDoctor**(EmployeeID, DoctorsAssocID, Name)
Primary key: EmployeeID
4. **CommissionedLeader**(EmployeeID, Name)
Primary key: EmployeeID
5. **Doctor**(EmployeeID, DoctorsAssocID, Name)
Primary key: EmployeeID
6. **Nurse**(EmployeeID, Name)
Primary key: EmployeeID
7. **SupportStaff**(EmployeeID, Name)
Primary key: EmployeeID
8. **Patient**(PatientID, Name, Age)
Primary key: PatientID
9. **Disease**(DiseaseID, Name)
Primary key: DiseaseID

Relationship relations (10):

10. **BelongsTo**(DepartmentID, HospitalID)
Primary key: DepartmentID

- Foreign keys: DepartmentID → Department(DepartmentID); HospitalID → Hospital(HospitalID)
11. **Has**(EmployeeID, DepartmentID)
Primary key: EmployeeID
Foreign keys: EmployeeID → ChiefDoctor(EmployeeID); DepartmentID → Department(DepartmentID)
 12. **Leads**(DepartmentID, EmployeeID)
Primary key: DepartmentID
Foreign keys: DepartmentID → Department(DepartmentID); EmployeeID → ChiefDoctor(EmployeeID) [UNIQUE]
 13. **LedBy**(EmployeeID, DepartmentID)
Primary key: EmployeeID
Foreign keys: EmployeeID → CommissionedLeader(EmployeeID); DepartmentID → Department(DepartmentID) [UNIQUE]
 14. **WorksAt_Doctor**(EmployeeID, DepartmentID)
Primary key: EmployeeID
Foreign keys: EmployeeID → Doctor(EmployeeID); DepartmentID → Department(DepartmentID)
 15. **WorksAt_Nurse**(EmployeeID, DepartmentID)
Primary key: EmployeeID
Foreign keys: EmployeeID → Nurse(EmployeeID); DepartmentID → Department(DepartmentID)
 16. **EmployedBy**(EmployeeID, HospitalID)
Primary key: EmployeeID
Foreign keys: EmployeeID → SupportStaff(EmployeeID); HospitalID → Hospital(HospitalID)
 17. **HeadBy**(HospitalID, EmployeeID)
Primary key: HospitalID
Foreign keys: HospitalID → Hospital(HospitalID); EmployeeID → Doctor(EmployeeID) [UNIQUE]
 18. **TreatedAt**(PatientID, DepartmentID)
Primary key: (PatientID, DepartmentID)
Foreign keys: PatientID → Patient(PatientID); DepartmentID → Department(DepartmentID)
 19. **DiagnosedWith**(PatientID, DiseaseID)
Primary key: (PatientID, DiseaseID)
Foreign keys: PatientID → Patient(PatientID); DiseaseID → Disease(DiseaseID)

Total: 19 relations.

Part (b): Minimal transformation (merge FKs into entity relations wherever possible)

Rule applied: only M:N relationships *must* stay as separate relations. Every 1:1 and 1:N relationship is folded into an entity table as a foreign key — placed on the mandatory (total-participation) side when the sides differ, to avoid unnecessary NULLs.

1. **Hospital**(HospitalID, Name, Address, HeadDoctorID)
Primary key: HospitalID
Foreign keys: HeadDoctorID → Doctor(EmployeeID) [NOT NULL, UNIQUE] (*from "head by"*)
2. **Department**(DepartmentID, Name, HospitalID, LeaderChiefDoctorID)
Primary key: DepartmentID
Foreign keys: HospitalID → Hospital(HospitalID) [NOT NULL] (*from "belongs to"*);
LeaderChiefDoctorID → ChiefDoctor(EmployeeID) [NULL, UNIQUE] (*from "leads"*)
3. **ChiefDoctor**(EmployeeID, DoctorsAssocID, Name, DepartmentID)
Primary key: EmployeeID
Foreign keys: DepartmentID → Department(DepartmentID) [NOT NULL] (*from "has"*)
4. **CommissionedLeader**(EmployeeID, Name, DepartmentID)
Primary key: EmployeeID
Foreign keys: DepartmentID → Department(DepartmentID) [NOT NULL, UNIQUE] (*from "led by"*)
5. **Doctor**(EmployeeID, DoctorsAssocID, Name, DepartmentID)
Primary key: EmployeeID
Foreign keys: DepartmentID → Department(DepartmentID) [NOT NULL] (*from "works at"*)
6. **Nurse**(EmployeeID, Name, DepartmentID)
Primary key: EmployeeID
Foreign keys: DepartmentID → Department(DepartmentID) [NOT NULL] (*from "works at"*)
7. **SupportStaff**(EmployeeID, Name, HospitalID)
Primary key: EmployeeID
Foreign keys: HospitalID → Hospital(HospitalID) [NOT NULL] (*from "employed by"*)
8. **Patient**(PatientID, Name, Age)
Primary key: PatientID
(*no foreign keys — treated_at and diagnosed_with are M:N and cannot be merged in*)
9. **Disease**(DiseaseID, Name)
Primary key: DiseaseID

Relations that must stay separate (M:N relationships only):

10. **TreatedAt**(PatientID, DepartmentID)
Primary key: (PatientID, DepartmentID)
Foreign keys: PatientID → Patient(PatientID); DepartmentID → Department(DepartmentID)
11. **DiagnosedWith**(PatientID, DiseaseID)
Primary key: (PatientID, DiseaseID)
Foreign keys: PatientID → Patient(PatientID); DiseaseID → Disease(DiseaseID)

Total: 11 relations (down from 19).

Question 5

Since every relationship in the schema is one-to-many (no M:N relationships exist), no separate relation is created for any relationship. Each relationship, along with its own attributes if any, is merged as a foreign key into the entity on the "many" (N) side. The foreign key is NOT NULL if that entity's participation has min = 1, and NULL-allowed if min = 0.

Relational Schema:

E1(**A11**, A12, A13, E3_1, AR4, E3_2)

E2(**A21**, A22, E1_1)

E3(**A31**)

E4(**A41**, E2_1, E3_1)

Foreign Key References:

- E1(E3_1) references E3(A31) — from R4
- E1(E3_2) references E3(A31) — from R5
- E2(E1_1) references E1(A11) — from R1
- E4(E2_1) references E2(A21) — from R2
- E4(E3_1) references E3(A31) — from R3

Keys and Null Analysis:

Relation E1(A11, A12, A13, E3_1, AR4, E3_2)

- Primary Key: A11
- A11 : NOT NULL
- A12 : NOT NULL
- A13 : NOT NULL
- E3_1 : NULL allowed (R4 participation of E1 is (0,1))
- AR4 : NULL allowed (attribute of R4; null exactly when the R4 link is absent)
- E3_2 : NOT NULL (R5 participation of E1 is (1,1))

Relation E2(A21, A22, E1_1)

- Primary Key: A21
- A21 : NOT NULL

- A22 : NOT NULL
- E1_1 : NOT NULL (R1 participation of E2 is (1,1))

Relation E3(A31)

- Primary Key: A31
- A31 : NOT NULL
- (No foreign key merges into E3, since E3 is always on the "N" side of R3, R4, and R5)

Relation E4(A41, E2_1, E3_1)

- Primary Key: A41
- A41 : NOT NULL
- E2_1 : NOT NULL (R2 participation of E4 is (1,1))
- E3_1 : NOT NULL (R3 participation of E4 is (1,1))

The only attributes that can ever be NULL are E3_1 and AR4 in relation E1, since R4 is the only relationship with (0,1) partial participation. All other foreign keys are NOT NULL because their corresponding participation constraints are (1,1).